

Direct activation of the ion channel TRPA1 by Ca^{2+} Sandra Zurborg^{1,2}, Brian Yurgionas^{1,2}, Julia A Jira¹,
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TRPA1 is an ion channel expressed by nociceptors and activated by irritant compounds such as mustard oil. The endogenous function of TRPA1 has remained unclear, a fact highlighted by ongoing debate over its potential role as a sensor of noxious cold. Here we show that intracellular Ca^{2+} activates human TRPA1 via an EF-hand domain and that cold sensitivity occurs indirectly (and nonphysiologically) through increased $[\text{Ca}^{2+}]_i$ during cooling in heterologous systems.

Intracellular Ca^{2+} is a key regulator of many transient receptor potential (TRP) channels, including TRPA1. Previous studies have shown that release of Ca^{2+} from intracellular stores can activate TRPA1 (ref. 1) and that responses to agonists such as mustard oil are substantially enhanced by increased $[\text{Ca}^{2+}]_i$ ^{1,2}.

We tested whether TRPA1 is directly activated by intracellular Ca^{2+} using patch clamping. In all experiments we used a tetracycline-inducible expression system to transiently express TRPA1 in HEK293 cells (Supplementary Methods online) and manipulated $[\text{Ca}^{2+}]_i$ by perfusion of buffered solutions through the patch pipette or directly onto excised patches. We observed robust activation of TRPA1 in response to elevations in $[\text{Ca}^{2+}]_i$ that was evident

immediately after establishment of whole-cell configuration or upon application of Ca^{2+} to inside-out patches (Fig. 1a,b). In whole-cell configuration, Ca^{2+} induced large inward and outward currents at -60 mV and $+80$ mV respectively that peaked within 50 s and then underwent pronounced inactivation back to baseline levels (Fig. 1a). The magnitude of the responses was dependent on $[\text{Ca}^{2+}]_i$ with a half-maximal effector concentration of 6 μM (Fig. 1c).

One interesting aspect of TRPA1 currents is their marked outward rectification at positive membrane potentials, which might reflect a voltage dependence of the channel (Fig. 1a inset). Recent evidence indicates that the voltage dependence of many TRP channels may be modulated by agonists, allowing for activation at physiological potentials³. We tested whether a similar mechanism might account for activation of TRPA1 by intracellular Ca^{2+} . In conditions nominally free of $[\text{Ca}^{2+}]_i$, currents could only be measured at potentials above $+60$ mV and the voltage for half-maximal activation ($V_{1/2}$) was 155 ± 25 mV. Increasing $[\text{Ca}^{2+}]_i$ induced a concentration-dependent leftward shift of the activation curve toward physiological potentials such that at 1 μM $[\text{Ca}^{2+}]_i$, the $V_{1/2}$ was 35 ± 9 mV, at 6 μM it was 14 ± 5 mV and at 100 μM it was -1 ± 3 mV (Fig. 1d and Supplementary Fig. 1 online). This indicates that voltage sensitivity and activation by intracellular Ca^{2+} are closely linked in TRPA1, an interaction that has

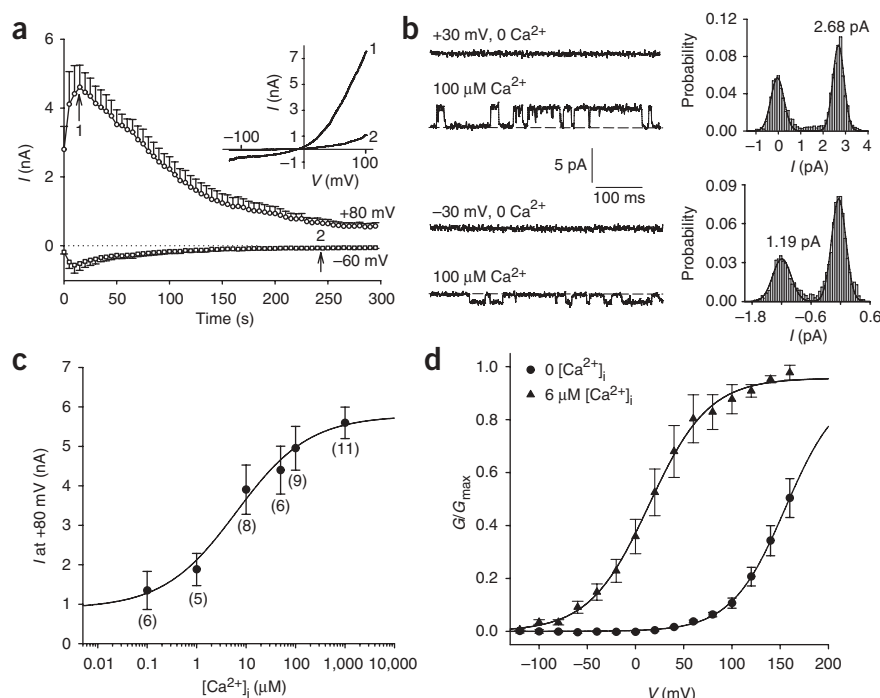


Figure 1 TRPA1 is activated by intracellular Ca^{2+} . (a) Average TRPA1 currents (I) at $+80$ mV and -60 mV in cells perfused with 1 mM Ca^{2+} ($n = 11$) and current versus voltage (V) curves of TRPA1 currents measured at times indicated (inset). (b) Single-channel recordings and amplitude histograms from inside-out patches at the indicated potentials. (c) Concentration-response curve of maximum TRPA1 currents measured at $+80$ mV. (d) Voltage dependent activation of TRPA1 in the absence of and at the half-maximal effector concentration for Ca^{2+} ($n = 9$ and 11). G/G_{max} , normalized conductance. Solid lines are best fits to Boltzmann functions. Data are expressed as mean $\pm$ s.e.m.

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Received 27 November 2006; accepted 2 January 2007; published online 28 January 2007; doi:10.1038/nn1843

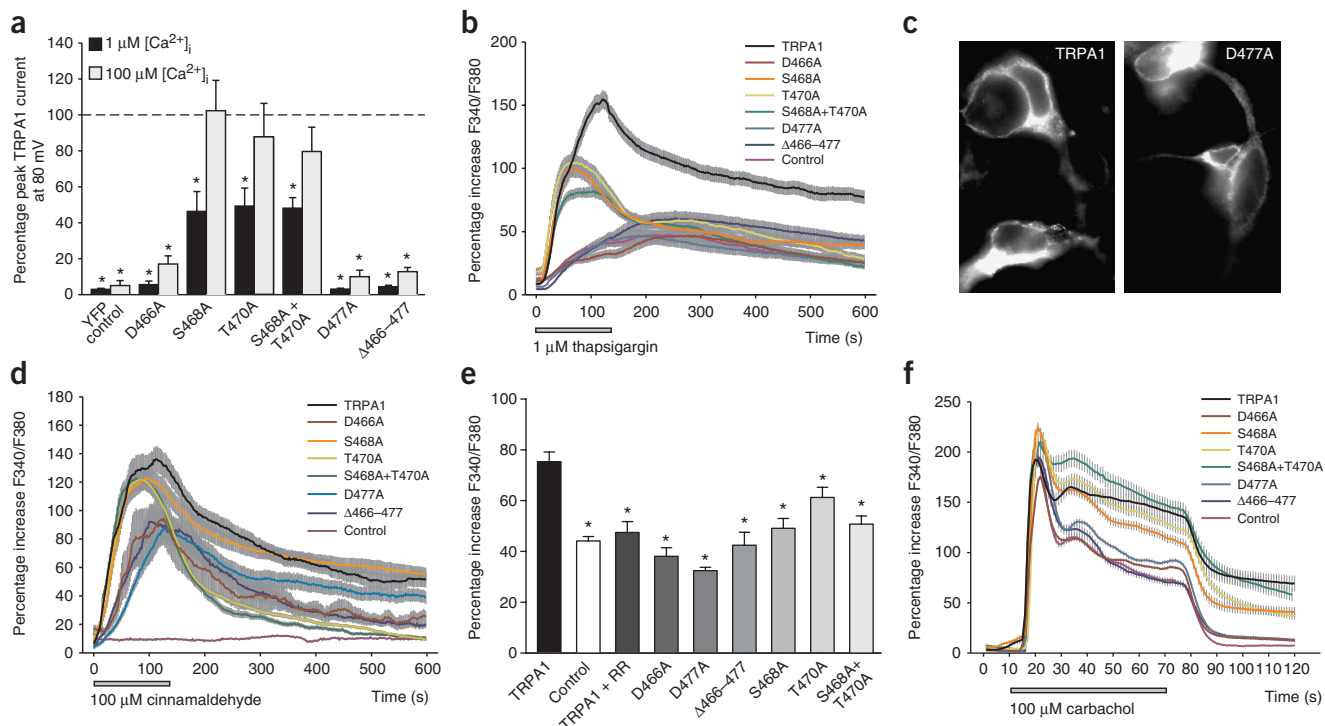


Figure 2 Mechanisms of Ca^{2+} activation of TRPA1. **(a)** Mutations within the EF-hand-like domain of TRPA1 reduce activation by Ca^{2+} . Percentage of control TRPA1 response (dotted line) at the indicated Ca^{2+} concentration ($n = 5\text{--}13$). **(b)** Mutations in the EF-hand-like domain reduce thapsigargin ($1\text{ }\mu\text{M}$)-evoked activation of TRPA1. F340/F380, ratio of fluorescence at 340 nm to that at 380 nm. **(c)** Expression of yellow fluorescent protein-tagged constructs in HEK293 cells. TRPA1 and D477A have similar distributions with substantial membrane expression. **(d)** Cinnamaldehyde ($100\text{ }\mu\text{M}$) activates TRPA1 and mutant channels. **(e)** Cold stimuli ($10\text{ }^{\circ}\text{C}$) elicit a rise in $[\text{Ca}^{2+}]_i$ in TRPA1-expressing and control cells. The TRPA1 response is reduced to control levels by $10\text{ }\mu\text{M}$ ruthenium red (RR) and by mutations in the EF-hand-like domain. **(f)** Carbachol ($100\text{ }\mu\text{M}$) evokes large sustained $[\text{Ca}^{2+}]_i$ responses in cells transfected with TRPA1. Sustained responses are significantly reduced in Ca^{2+} -insensitive TRPA1 mutants. $*P < 0.05$ compared to TRPA1-transfected cells.

also been reported for TRPM4 (ref. 4) and TRPM5 (ref. 5), other Ca^{2+} -activated TRP channels.

What is the mechanism of Ca^{2+} activation of TRPA1? Calmodulin (CaM) is a ubiquitously expressed Ca^{2+} -binding protein that is an important mediator of Ca^{2+} -dependent regulation of TRP channels⁶. We tested whether CaM might be responsible for Ca^{2+} activation of TRPA1 by coexpressing a dominant negative CaM mutant unable to bind Ca^{2+} (CaM_{1,2,3,4}) with TRPA1. As a positive control, we also examined CaM_{1,2,3,4} modulation of the related TRP channel TRPM4 (ref. 7). At all intracellular Ca^{2+} concentrations tested, CaM_{1,2,3,4} had no effect on Ca^{2+} -activated TRPA1 currents, but did inhibit TRPM4 currents (**Supplementary Fig. 2** online). Similarly, in pharmacological experiments, intracellular application of either wild-type CaM or a CaM antagonist (W-7, Calbiochem) did not affect Ca^{2+} activation of TRPA1 (**Supplementary Fig. 2**). Thus, even though CaM is essential for the function of many other TRP channels, it does not appear to regulate TRPA1.

In the absence of TRPA1 modulation by CaM, possible mechanisms for Ca^{2+} activation include interaction with other Ca^{2+} binding proteins or direct gating by an intrinsic Ca^{2+} sensor. Intriguingly, TRPA1 is annotated as containing a putative Ca^{2+} -binding EF-hand domain in its N terminus. We therefore generated mutations within this domain by deleting either the putative Ca^{2+} -binding loop ($\Delta 466\text{--}477$) or by substituting alanines for key residues at D466, S468, T470 and D477. We then used whole-cell patch clamping to assay Ca^{2+} sensitivity. Notably, we found that the $\Delta 466\text{--}477$, D466A and D477A mutants were completely insensitive to increases in $[\text{Ca}^{2+}]_i$, and the

S468A, T470A and double S468A+T470A mutants had significantly reduced sensitivity at low Ca^{2+} concentrations (**Fig. 2a**).

We next examined Ca^{2+} activation of these mutants in intact cells. We treated cells with thapsigargin to induce release of Ca^{2+} from intracellular stores and measured channel activity using Ca^{2+} imaging. In agreement with our patch clamping data, thapsigargin-evoked Ca^{2+} release induced a robust response in TRPA1-expressing cells. Moreover, this response was abolished in cells expressing the $\Delta 466\text{--}477$, D466A and D477A mutant channels and markedly reduced by the S468A, T470A and S468A+T470A mutations (**Fig. 2b**).

It is possible that mutation of the EF-hand-like domain in TRPA1 disrupts membrane targeting of channels or renders them nonfunctional. We therefore examined the cellular distribution of mutant channels and measured their activation by other agonists. Yellow fluorescent protein-tagged fusion proteins of TRPA1 mutants had a significant proportion of protein visible in the membrane and all showed very similar distributions to that of wild-type TRPA1 (**Fig. 2c**). Furthermore, all mutant channels responded to the TRPA1 agonists cinnamaldehyde (**Fig. 2d**) and mustard oil (data not shown). These responses were delayed and reduced in mutant channels, but as Ca^{2+} is required for full and rapid activation of TRPA1 by mustard oil^{1,2}, a reduced responsiveness of Ca^{2+} -insensitive mutants is consistent with previous reports.

We have demonstrated that the putative EF-hand of TRPA1 is required for activation of TRPA1 by Ca^{2+} , but whether this region binds Ca^{2+} in the same way as a classical EF-hand domain is not clear. EF-hand domains normally occur in adjacent pairs⁸ and we found no

evidence of a second EF-hand domain in TRPA1. As TRP channels are believed to form tetrameric structures⁹, EF-hands might pair in this region by oligomerization, as has been proposed for the protein calpain⁸. However, in the absence of structural information, it is not possible to determine the details of Ca²⁺ binding here. Nonetheless, our functional data do demonstrate that disruption of key residues in this region strongly decreases Ca²⁺ sensitivity of TRPA1.

We next examined how the Ca²⁺ sensitivity of TRPA1 might be important for proposed physiological roles of the channel. One function of TRPA1 that has attracted much attention is its potential role as a sensor for painful cold. Despite several cellular^{1,2,10} and two different mouse knockout studies^{11,12}, there is still no clear consensus as to whether TRPA1 is directly activated by cold. We investigated this in our system using ratiometric Ca²⁺ imaging. We observed that cells expressing TRPA1 did indeed show a substantial response to cooling (Fig. 2e). However, we also saw significant cold responses in control cells not expressing TRPA1. Both of these responses had remarkably similar activation thresholds (TRPA1 17.8 ± 0.6 °C and control 17.2 ± 0.4 °C), which led us to question whether TRPA1 might be activated by increases in [Ca²⁺]_i during cooling rather than directly by cold temperatures. We tested this by performing pharmacological experiments and by examining the response of Ca²⁺-insensitive mutant channels to cold. We found that the TRP channel blocker ruthenium red inhibited cold responses in TRPA1-containing cells, but only to control levels. Likewise, cells expressing Ca²⁺-insensitive mutants showed cold sensitivity similar to that seen in control cells (Fig. 2e).

Our data implied that TRPA1 was indeed activated by elevation of [Ca²⁺]_i during cooling rather than directly by cold. We explored this further using whole-cell patch clamping in which we buffered [Ca²⁺]_i at a constant level. In Ca²⁺-free conditions we never observed currents or changes in the voltage dependence of TRPA1 in response to cold (Supplementary Fig. 3). However, Ca²⁺ could still be essential for cold sensitivity of TRPA1, as has been reported for heat activation of TRPM5 (ref. 13). We tested this by applying cold stimuli at 1 μM [Ca²⁺]_i and found that cooling did not evoke TRPA1 currents but actually reduced Ca²⁺ activation of TRPA1, as would be expected for a channel that is not cold sensitive (Supplementary Fig. 3). Collectively these data indicate that TRPA1 is not directly gated by cold but is activated by increased basal [Ca²⁺]_i in HEK293 cells during cooling. As sensory neurons do not show indiscriminant Ca²⁺ fluxes in response to cooling¹⁴, TRPA1 is unlikely to function as a cold sensor *in vivo*.

Conceivably, any signal leading to an increase in [Ca²⁺]_i in TRPA1-expressing sensory neurons could activate the channel. Several studies at the cellular and whole-animal levels have demonstrated a robust coupling between inflammatory mediators that stimulate phospholipase C (by way of G protein-coupled receptors) and TRPA1 (refs. 1,11,15). We tested the role of Ca²⁺ in this context using

ratiometric Ca²⁺ imaging. Application of the cholinergic agonist carbachol to HEK293 cells evoked a transient increase in [Ca²⁺]_i followed by a sustained elevation that was significantly potentiated in the presence of TRPA1. The sustained response was also enhanced in the S468A, T470A and S468A+T470A mutant channels but not in the D466A, D477A or Δ466–477 mutants (Fig. 2f). This would indicate that large increases in [Ca²⁺]_i induced by carbachol outweigh the reduced Ca²⁺ sensitivity of S468A, T470A and S468A+T470A mutant channels at low [Ca²⁺]_i. Furthermore, because the completely Ca²⁺-insensitive mutants (D466A, D477A and Δ466–477) did not show an elevated response to carbachol, it also indicates that Ca²⁺ is essential for activation of TRPA1 downstream of phospholipase C.

The present findings demonstrate that intracellular Ca²⁺ is an important endogenous ligand of TRPA1. Moreover, the identification of the Ca²⁺ activation domain in TRPA1 could serve as a starting point for the development of TRPA1 antagonists with prospective clinical applications as new types of analgesics.

Note: Supplementary information is available on the Nature Neuroscience website.

ACKNOWLEDGMENTS

We thank J.P. Adelman (Oregon Health & Science University) and S.E. Jordt (Yale University) for providing CaM_{1,2,3,4} and human TRPA1 clones, respectively, and C. Stein and G. Lewin for comments. Supported by Deutsche Forschungsgemeinschaft grants HE-4483 and KFO100.

AUTHOR CONTRIBUTIONS

S.Z. and B.Y. conducted the electrophysiology and Ca²⁺ imaging experiments. J.A.J. conducted the CaM experiments, O.C. made the cell line, and S.Z., B.Y., J.A.J. and P.A.H. analyzed the data. P.A.H. wrote the manuscript.

COMPETING INTERESTS STATEMENT

The authors declare that they have no competing financial interests.

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